

ANDRES SANTIAGO DUCUARA VELASQUEZ

Website ◇ 

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EDUCATION

Graduate

- **Computer Science & Artificial Intelligence** **Universidad Sergio Arboleda**
Engineer in Computer Science and Artificial Intelligence April 2024

Masters

- **Master in Artificial Intelligence** **Universidad Sergio Arboleda**
April 2026

EXPERIENCE

Universidad Sergio Arboleda

February 2025 – Present

Professor

Bogotá, COL

- In charge of the course "Algorithmic Thinking," which covers introductory engineering topics such as logic, problem-solving, and the fundamentals of programming.
- In charge of the course "Cybernetics and Intelligent Systems," which explores the core principles of cybernetics, the early developments of artificial intelligence with McCulloch-Pitts, and complex systems such as chaos theory.

Contraloría General de la República

August 2024 – December 2024

Data Engineer

Bogotá, COL

- Structuring and continuous improvement of analytical models through efficient automation of each stage of the ETL process.
- Implementation of solutions based on Azure and Databricks, using Python and SQL to optimize the processing and analysis of large data volumes.
- Design and deployment of interactive dashboards in Power BI for effective visualization of analytical results.

Unidad de Planeación Minero Energética - UPME

May 2024 – December 2024

Artificial Intelligence Engineer

Bogotá, COL

- Development of pilot projects and research of artificial intelligence solutions to optimize critical processes within the organization.
- Integration of artificial intelligence solutions.

Contraloría General de la República

November 2023 – December 2023

Data Analyst

Bogotá, COL

- Provision of services to support document management for the Information Unit of the Directorate of Information, Analysis, and Immediate Reaction (DIARI).
- Support in various projects of the entity related to OCR and web applications on Azure cloud architecture.

Universidad Sergio Arboleda

August 2023 – June 2024

Mathematical talent's teacher

Bogotá, COL

- Teaching courses related to technology, such as robotics, communication protocols, programming fundamentals, among others, aimed at children and adolescents.

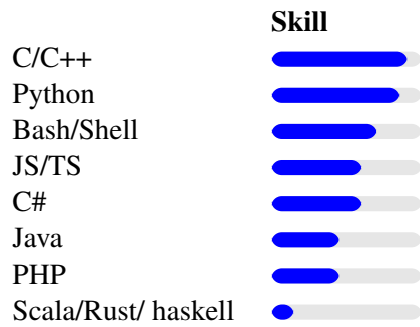
- Provision of management support services in the administrative assistance, handling of information and documentation of the activities of the DIARI. (Dirección de información, Análisis y Reacción Inmediata).
- Structuring public data from territorial entities at the central level and universities, implementing Azure services (Pipelines and Databricks).
- Development projects related to the use of OCR, web scraping, Python, or Azure applications.

CERTIFICATES

- Fundamentals of Accelerated Computing with CUDA C/C++ [NVIDIA DLI]
- NDG Linux Essentials [NDG Netdevgroup]
- Introduction to Packet tracer [CISCO Networking Academy]
- Programming Essentials in Python [EDG Python Institute]
- Programming Essentials in C++ [EDG C++ Institute]
- Database Design and Programming with SQL – Oracle [Oracle Academy]
- Programación en Java [Politécnico de Colombia]
- Programación en PHP [Politécnico de Colombia]

TECHNICAL STRENGTHS

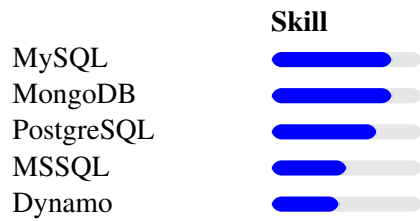
Computer Languages



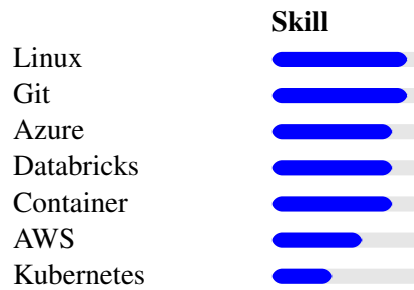
Frameworks



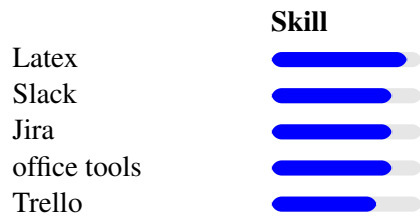
Databases



Cloud & DevOps skills



Productivity & collaboration suites



High performance & AI computing

- **ML:** TensorFlow, PyTorch, Keras, Scikit-learn
- **GPU Computing:** CUDA, Numba, OpenCL
- **NLP:** Hugging Face, Transformers, LangChain
- **Parallel Computing:** MPI, OpenMP, SYCL
- **Distributed Systems:** Apache Spark, Hadoop
- **Electronics:** Embedded Systems, IoT, Robotics, EDA